Knowledge Base Entry — Glostrup Hospital

Functional Imaging Unit & Clinic for Clinical Physiology, Nuclear Medicine and PET

Role Overview

Biomedical Engineering Student Assistant / Research Assistant and Engineering Intern within a highly regulated hospital environment, supporting clinical research and advanced medical imaging across MRI, CT, PET, SPECT, and nuclear medicine modalities. Operated at the intersection of patient-facing clinical work, medical device quality assurance, and data-driven research, with a strong emphasis on patient safety, protocol standardization, and analytical rigor.

Clinical Imaging & Patient-Facing Responsibilities

- Performed **independent MRI scanning** (always dual-staffed for safety compliance) on a **Philips 3T system**, primarily supporting renal functional MRI and flow-based imaging in clinical research trials.
- Prepared and received patients for scans, including:
    - Comprehensive **MR safety screening** (metal objects, implants, tattoos, contraindications).
    - Patient instruction, reassurance, and monitoring, including managing claustrophobia-related distress and terminating scans when required for patient wellbeing.
- Configured and applied **physiological monitoring equipment**, including:
    - 4-point ECG, pulse oximetry, and blood pressure monitoring.
    - Continuous assessment and mitigation of motion, contact, and signal artefacts to ensure diagnostic-quality data.
- Executed predefined research protocols to ensure **data equivalence across all subjects**, while adjusting slice positioning and anatomical coverage to achieve accurate organ representation.
- Managed post-scan workflows, including secure transfer of imaging data to project repositories and structured research folders.

Data Analysis, Image Processing & Research Support

- Conducted post-processing and quantitative analysis of MRI data, including **renal perfusion and flow metrics**, using **MATLAB**, FSL, and custom in-house analysis software.
- Annotated and curated research datasets to support blinded clinical trials and PhD-level research projects.
- Identified systematic sources of error and variability in imaging-based perfusion analysis, contributing directly to improved research reliability.
- Authored a bachelor thesis based on observed analytical bottlenecks, developing a **deep learning–based segmentation and co-registration pipeline** for renal MRI time series:
    - Automated kidney segmentation and alignment across 50+ image series per scan.

    - Significantly reduced manual workload and uncertainty in perfusion calculations.
    - Model was later implemented into the department's analysis workflow, improving robustness of clinical trial outcomes.

Medical Device Quality Assurance & Technical Operations

- Executed **hands-on quality assurance programs** for a broad range of diagnostic equipment, including:
    - MRI, CT, PET-CT, SPECT, gamma cameras, DEXA, ultrasound, lung function systems, dosimetry equipment, and isotope handling infrastructure.
- Performed daily, weekly, monthly, and annual controls using phantom-based and non-phantom-based procedures to ensure regulatory compliance.
- Documented deviations, evaluated tolerance thresholds, initiated corrective actions where possible, and escalated findings to senior physicists and engineers.
- Detected and monitored imaging artefacts, including CT system deviations that required longitudinal surveillance rather than immediate intervention.
- Diagnosed failures in **contrast and isotope injection pumps**, successfully repairing one device and coordinating vendor repair for another, recognizing the **patient-critical risk** associated with malfunctioning infusion equipment in nuclear medicine.

Protocol Standardization, SOPs & Systems Thinking

- Led a cross-scanner **CT protocol harmonization effort** across approximately a dozen systems with varying hardware generations:
    - Audited protocol parameters across all scanners.
    - Standardized settings to ensure that identical protocols produced equivalent examinations regardless of scanner or patient relocation.
    - Documented unavoidable deviations due to hardware constraints and ensured traceability.
- Authored **standard operating procedures (SOPs)** for clinical staff to support infrequent but critical workflows (e.g., scan recreation and protocol recovery), reducing reliance on ad hoc expert assistance.
- Updated departmental safety and technical documentation to reflect current software behavior and operational practices.

Professional Attributes & Differentiators

- Demonstrated **exceptional trustworthiness and autonomy**, routinely operating MRI scans and handling patients during weekends with minimal on-site supervision.
- Known for **precision, attention to detail, and accountability for patient safety** in high-stakes clinical environments.

- Strong problem solver with a systems-level mindset, proactively identifying inefficiencies and proposing technically sound improvements.
- Experienced in interdisciplinary collaboration with physicians, physicists, engineers, radiographers, bioanalysts, and academic researchers.
- Developed a nuanced understanding of the practical and ethical complexities inherent in clinical trials, including variability control, error propagation, and patient risk management.

Core Keyword Clusters (for GPT Retrieval)

Medical Imaging: MRI, CT, PET, SPECT, Nuclear Medicine, Renal fMRI, Perfusion Imaging, Flow Measurements
Clinical Engineering: Medical Device QA, Phantom Testing, Protocol Harmonization, SOP Development, Regulatory Compliance
Data & AI: MATLAB, Image Analysis, Medical Image Processing, Deep Learning, Segmentation, Co-registration, Data Quality
Clinical Research: Clinical Trials, Blinded Studies, Patient Safety, Research Data Pipelines
Professional Skills: Risk Assessment, Precision Engineering, Interdisciplinary Collaboration, Technical Documentation

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Engineering Insight & Impact (Job-Focused Reasoning)

- Learned firsthand how **human variability** (operator fatigue, time pressure, contextual bias) can introduce subtle but significant inconsistencies in medical image annotation and analysis.
- Identified that manual kidney alignment across MRI time series was a major, under-recognized source of uncertainty in renal perfusion measurements.
- Proposed and developed an **automation-driven solution** using deep learning to segment and co-register kidneys consistently across image series.
- The automated pipeline **reduced re-analysis**, improved reproducibility of perfusion metrics, and increased overall confidence in clinical trial results.
- Gained a strong appreciation for **automation as a quality and risk-mitigation tool**, rather than merely a productivity enhancement.

Reusable Professional Insights

- “During my time at Glostrup Hospital, I learned how even small automation steps can significantly improve data accuracy and reproducibility in clinical research.”
- “Removing human variability from a critical annotation step reduced hidden sources of error and the need for repeated analyses.”
- “Automation, when applied selectively, can act as a safety and quality control mechanism in regulated environments.”